
When Do Language Servers Help Coding Agents?

Ian Barber

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Abstract

Coding agents are increasingly given language-server tooling — go-to-definition, compact definition spans, type diagnostics — on the assumption that better code intelligence makes a better agent. We tested that against a capable text baseline of grep, ranged reads and a shell, using seven models over constructed dispatch and seeded-defect suites, retrieval tasks built on real library source, and a shell agent on real repositories.

Availability alone rarely paid. Go-to-definition was token-neutral wherever the receiver’s type was readable in source the agent already opens, because the model reads the type and localizes itself; it lifted resolution only where the type was withheld, and then only when the agent was prompted to use it. A compact span — from a static definition tool, not a live server — cut total tokens 3.3–4.1x against whole-file retrieval, and 1.3x against grep plus ranged reads in the one suite where it genuinely replaced the read; elsewhere models read the file anyway, on 35 of 36 pushed spans. A type checker lifted accepted correct outcomes from 1/12 to 11/12 at the submission boundary and changed nothing during editing, on a channel the model exercised once in twelve.

What moved outcomes in every thread was a change in the agent’s policy, not the presence of the tool. Thirty-nine demonstrations taught a 27B to stop rereading, and the learned policy proved conditional rather than blanket: where the span genuinely lacked the defect it went looking every time. Installing the server is not the intervention.

1. Introduction

A language server answers questions about code. So does a coding agent holding grep, ranged reads and a shell, and it answers most of them for itself. What the server adds therefore turns on three narrower questions: whether it answers something the agent cannot, whether it answers in a cheaper form, and whether it answers at a better moment.

IF: whether delivering the semantic information changes what the agent finds. A yes is the semantic arm resolving targets the text arm misses; a no is both arms resolving the same tasks at the same cost.

FORM: whether a compact span costs less than reading the file. A yes is total tokens falling at unchanged success; a no is the span arriving as context on top of the reads it was meant to replace.

WHEN: whether the moment a diagnostic arrives changes the outcome. A yes is the same diagnostic changing the accepted result depending only on when it fires; a no is the outcome flat across delivery phases.

Each is a claim tool builders make. Typed Holes [1] and LSPRAG [2] push types and definitions into the context on the premise that the information is missing; the first question tests that premise. STALL+ [3] reports static-analysis value varying with integration phase; the third tests that. Neither line reports a counterfactual that already reads code well, so the baseline here is a capable text agent.

2. How it was tested

Every result but one comes from a purpose-built harness: a model drives a fixed six-action loop — grep, ranged read, whole-file read, line edit, run tests, submit — over a Python workspace, with one semantic operation added or withheld per arm; the exception is one out-of-harness probe on real repositories. Nothing ran through an MCP server, a skill or an editor plugin.

Dispatch: fifteen tasks where a typed receiver calls one of about ten same-named overrides, one buggy, across a visibility ladder (L0 call-site annotation, L1 test construction site, L2 factory indirection) and a hidden rung withholding `app.py` and the test source, receiver construction redacted. `effic` and `effic_real2`: retrieval-cost tasks, `effic_real2` constructed misuse over vendored real library source (toolz 1.1.0, more-itertools 11.1.0). navigation-v2: mechanically validated definition-span instances. checker-gate-v3: twelve seeded defect/clean pairs, each defect coherent, visible-passing and held-out-failing with one target-scoped semantic diagnostic, each clean control its validated gold. The other suites, `effic` included, are constructed from scratch: the real-repository scan cleared no admissible candidate (C17).

The efficiency thread’s definition operation is a static AST resolver (`skip_pyrefly=True`, `ast.parse`), not a live language server; live Pyrefly ran only in dispatch goto arms and one 7B run.

Temperature 0, one rollout per cell: the dispatch ladder, navigation-v2, the paired retrieval run and checker-gate-v3. Temperature 0.7, repeated seeds: the hidden rung and its visible control (five seeds per task, 450 rollouts), the whole-file contrast, the election and execution-feedback runs (four seeds behind the 44-attempt cells), and the substitution harvest, with its held-out retest at 0. Models: Qwen2.5-Coder-7B, Qwen3.5-27B, Qwen3.6-27B locally; Sonnet 4.5, DeepSeek v3.1, GLM-4.6, GPT-5.6 “Luna” via API. Unit of analysis: the task, with task-level bootstrap intervals.

Per-claim configuration and artifacts: [ledger](#), [manifest](#), [protocols](#).

3. Results

3.1 Does delivering the semantic information change what the agent finds?

On the annotated rung, `grep` and ranged reads resolved 15/15, neutral goto 14/15 and framed goto 15/15, at matched-success ratios 0.972 and 1.041 (C9). Across L0, L1 and L2, mean input tokens on resolved tasks were 1,436, 1,429 and 1,465 at 14–15/15, goto ratios 0.945–1.065 at every rung including L2, where only the server statically resolves the type. Trajectory counters: about 0.3 greps per task, the receiver’s type read where it appeared, the buggy override opened first.

The hidden rung, against a matched visible control (C30):

Arm	Hidden	Paired difference vs grep	Greps	Whole-file reads	Visible
<code>grep_base</code>	70/75	—	1.52	2.17	75/75
<code>defn_avail</code>	71/75	+0.013 [−0.040, +0.067]	1.47	2.04	75/75
<code>defn_prompt</code>	75/75	+0.067 [+0.013, +0.133]	0.36	0.68	75/75

Per-task resolution rates over five seeds at temperature 0.7, 95% task bootstrap intervals over the fifteen tasks; every arm resolved every rollout in the visible control, so the effect appears only where the type was withheld.

Framed lookup improved four tasks (`codec_serialize`, `job_priority`, `resource_cost`, `row_format_row`) and worsened none; the same operation merely available improved three and worsened two. Cost landed on resolution, not tokens: matched-success input-token ratios were 0.983 framed and 1.005 available. The constructed hidden regime withholds source a real agent could usually open.

Typed lookup mechanically reached the exact gold override among 8–15 same-named overrides where erased lookup returned a non-discriminating base, both variants type-clean (C15). The two-task agent arm passed all 12 task-condition cells at a typed automatic/baseline total-token ratio of 1.037 (0.988–1.093), too wide for equivalence; every automatic result was followed by a target-file read, and composed lookup added about six seconds per task (C24).

3.2 Does a compact span cost less than reading the file?

On the retrieval-cost tasks, compact spans cut total tokens 3.35x, 3.49x and 4.12x against whole-file retrieval for Qwen3.6-27B, Sonnet 4.5 and DeepSeek v3.1, 44/44 resolved in every arm (C1).

Against `grep` plus ranged reads with whole-file fallback, eleven `effic_real2` tasks with Qwen3.5-27B cost 1,602 mean total tokens with text against 1,235 with definitions, a paired ratio of 1.297 (task bootstrap 1.093–1.527), cheaper on 10/11 at 11/11 success in both arms, the one suite and model where the span replaced the read, with 0 of 11 definitions followed by a reread (C27).

Untrained, the 7B elected the definition operation in 1 of 48 rollouts; relabel-trained, 48 of 48 with no file reads, mean input 3,086 to 687, resolution 31/48 to 48/48 (C2). Framing raised codenav use on two of three real-repository tasks at one seed (C3, C6).

With `lsp_defn` on in both `effic` arms, the untrained 7B invoked the definition tool 0/24 and the trained arm 24/24, mean input 2,894 to 689, success 14/24 to 24/24. The adapter is the election mechanism. A live-first Pyrefly backend with historical AST fallback reproduces the saving once the policy elects, backend unrecorded per row. No arm held the policy fixed while varying the backend, so this is not a measurement of a language-server token effect (C4).

Pushed spans were reread 35/36, and 34/36 with an explicit sufficiency instruction that removed 2/36 and added 294 mean tokens for the 27B (C31).

Model	Reread, pushed span	Reread, plus sufficiency instruction
Qwen3.6-27B	11/12	12/12
Sonnet 4.5	12/12	12/12
DeepSeek v3.1	12/12	10/12

A separate run with Qwen3.6-27B, GLM-4.6 and GPT-5.6 Luna reproduced the pushed reread at 35/36. Among instances each model elected, reread was 9/9, 4/9 and 7/12; GLM-4.6 solved only 5–6/12 of these tasks, and reread counted independently of success (C34).

In a bash agent (mini-swe-agent, Sonnet 4.5) on real repositories, roughly 16 of 18 `codenav defn` calls were followed by a read of the file just resolved, `sympy's str.py` at 22 `sed` invocations. Three tasks, one seed, 60-step cap, from committed summaries. That agent took 0–3 whole-file reads in 44–60 actions (C6, log).

A DAGger-style relabel [4] run on 39 demonstrations, none read-required, cut the post-span reread 11/12 to 0/12 on held-out instances disjoint in seed and template, mean input 1,157 to 748, a matched-success saving of 1.59x. Held-out pass moved 11/12 to 10/12: one rescue against two `xor` losses that pass visible and fail held-out, not distinguishable from noise at $n=12$ and one seed (C33). A pre-registered confirmation on twelve reserved instances, disjoint from the harvest in both seeds and templates and never run before, reproduced it: reread 12/12 untrained against 1/12 trained (11 removed, 1 persisted, 0 induced), mean input 1,223 to 729, a matched-success saving of 1.725x, and held-out pass 12/12 to 11/12 (C35).

What the training produced is a conditional policy, not suppression. Twelve further instances hoist each override's arithmetic into a module-level helper, so the span live Pyrefly returns for the call site is the complete definition of the method that binds and the bug sits outside it; a byte-paired twin differing in one line restores sufficiency, and a third arm leaves the repo identical and delivers a second genuine lookup at the helper (C36):

Reads after span	Span insufficient	Span sufficient	Helper also delivered
Untrained	12/12	12/12	11/12
Trained	12/12	2/12	0/12

Both conditions repaired 12/12 in every arm, visible and held-out, so this is retrieval behaviour at fixed correctness. Where reading was required the trained model read a mean of 1.00 times against the untrained model's 3.42, and where it was not, 0.17 and 0.00 against 1.58 and 1.75; input tokens were lower in every arm. Two apparatus gates matter: an adversary over seventeen one-parameter repair forms reached held-out on none, and the harness's post-edit file view was redacted, since it would otherwise have delivered the hidden line with no read.

3.3 Does the moment the diagnostic arrives change the outcome?

Accepted type-clean and held-out-correct outcomes were 1/12 under control, 10/12 with one-shot diagnostics at revision and 11/12 behind a submission gate, for task-bootstrap effects against control of +0.375 [+0.250,

+0.500] and +0.417 [+0.292, +0.500]. After-every-edit stayed at 1/12, +0.000 [−0.125, +0.125], on a channel that fired in 1 of 12 defect rows because the model edited before submitting in only 1 of 12 (C32).

Twelve seeded defect/clean pairs	Control	After every edit	One-shot	Gate
Bad completion accepted	11/12	11/12	2/12	1/12
Revision tokens, defect	787	754	1,210	1,380
Wall time, defect (s)	27.8	14.5	68.3	62.7
Revision tokens, clean	591	—	619	591
Wall time, clean (s)	9.4	9.9	9.6	9.5

After-every-edit fired in 1 of 12 defect rows; mean edits 0.08 against 0.17 in control.

The gate matched control on clean work in tokens and wall time, and spent about 2.2x control’s wall time on defective work. It rejected 10 of 12 submissions, all 10 repairing, retesting and resubmitting, and 2 of 12 self-repaired beforehand; on `auth_pipeline_handler` the model received the diagnostic and submitted anyway with zero edits, so the gate’s one-task edge over revision came from refusal, not information. Its single miss, `auth_shapes_protocol`, was type-clean after self-repair and failed held-out. The 0/12 false rejections are an apparatus check, since each clean control is that defect’s validated gold and the checker is deterministic; population false-rejection rate is unmeasured. Calibration produced no usable natural checker-detectable defect: frontier inference sat at 18/18 and a descriptive 27B authoring arm at 12/12, 0/3 natural 7B drafts were coherent, and 2/8 coherent 14B drafts were already type-clean, so these twelve defects were seeded (C10, C11, C16, C22). A separate execution-feedback grid (two frontier models, 14 tasks, three seeds, three delivery modes) passed 252/252 (C12).

4. Findings

4.1 Delivery

On a fifteen-task constructed dispatch suite at a single seed, one rollout per cell, semantic navigation added little over grep and ranged reads: a capable model reads the receiver’s type wherever it is visible and opens the right file itself, so the lookup delivers what the agent already holds. Cost stayed flat as the type moved from a call-site annotation to the test’s construction site to factory indirection: what carries resolution is correct types in the source, not the server that queries them. The exception is a constructed regime withholding the type-bearing source, where framed lookup resolved every rollout over fifteen tasks at five seeds and text did not; the same operation merely available changed nothing, so what the server adds is contingent on election. Sound types also sharpen the resolver mechanically, but that precision bought no agent-level outcome in a two-task pilot.

4.2 Form

A compact span is cheaper than a read only when it replaces the read, and whether it does is a property of the agent’s policy, not the tool. The span is cheaper by a wide margin against whole-file retrieval and by a narrower one against competent grep plus ranged reads. Election is separable: framing changed which operation the capable models called, and training changed it on a 7B. Substitution did not follow: on twelve instances per model at one seed, a pushed span was reread by a local 27B and by four frontier API models in two independent runs, and neither a sufficiency instruction nor self-election removed it. In a three-task, single-seed probe with a frontier model on sympy, astropy and sphinx, definition calls sat on top of reads; the whole-file read the largest ratios are measured against was rare there, so the achievable regime is the smaller one. Relabel training removed the reread where instruction and election could not, on one model at one seed and again on a pre-registered confirmation set built from templates the training never saw. What it trained is conditional rather than blanket. Where the delivered span was genuine but did not contain the defect the trained model read every time, and once rather than the untrained model’s three times; where the span sufficed it read twice in twelve, and where the missing definition was itself delivered, never. Correctness sat at ceiling throughout, so training changed when the agent retrieves rather than whether it succeeds, and it learned the distinction from demonstrations none of which required a read.

4.3 Timing

On twelve seeded defect/clean pairs, one model, one seed, a checker at the submission boundary converted accepted bad completions into repaired ones and one-shot delivery at revision performed comparably, while delivering the same checker after every edit changed nothing, on a channel that fired in one of twelve defect rows. Late delivery works; early delivery is unmeasured. The submission gate is preferable to delivery at revision because it charges its cost only to defective submissions: clean work cost no more than control in tokens or wall time, and the price fell on wall time when a submission was defective. A type-clean gate is only as behaviorally sound as its checker. Natural prevalence is unmeasured, and the same ceiling that forced seeding left execution feedback nothing to move on the tested small-task suite.

5. If you use a coding agent

On a typed Python codebase, the clearest return is a type check at the submission boundary with a repair-and-resubmit loop.

Lever	Do this	Changes if
Do nothing	Add no semantic navigation tool over typed source the agent already opens (C9, C30)	Your source hides receiver types
Add a tool or MCP server	Add one only where source the agent opens hides the receiver’s type — generated stubs, vendored or compiled boundaries — and frame its use (C30, C27)	You measure targets the agent misses
Skill or prompt change	Tell the agent which operation to use and when, and verify post-span reads before budgeting a saving (C3, C30, C31, C34)	A prompt removes the post-span read
Wire a hook	Gate submission on the checker, requiring repair and retest before acceptance; after-every-edit firing is untested (C32)	Your checker covers behavior, or your model edits repeatedly
Fine-tune	Fine-tune only with weights you control; it buys a conditional policy — the trained model still read every time the span did not contain the defect, and once rather than three times (C33, C36)	Your insufficiency looks unlike a delegation

Keep type annotations correct and present; the text baseline exploited exactly that. On your own workload, measure wrong-file edits, post-span reads of the defining file and spurious gate rejections; the clean-work cost here came from workspaces guaranteed clean. No result here ranks MCP servers, skill files and project instructions as delivery surfaces.

6. Future work

The line most worth following is retrieval calibration. Thirty-nine demonstrations, none of which required a read, produced a model that goes looking when what it holds is inadequate and stops when it is not — a judgment about the sufficiency of retrieved context, learned from examples that never exercised it. Nothing in that is specific to definition spans. Agents retrieve constantly and mostly indiscriminately: grep hits, file chunks, documentation, search results, the output of other tools. If “do I have enough?” is trainable rather than a property of one tool, it is a general lever on agent cost, and the open questions are general ones. Does the judgment transfer across retrieval channels, or was it learned about one? Does it survive forms of inadequacy structurally unlike the delegation tested here, or is the model reading a surface cue rather than testing sufficiency? Does it hold at other scales, outside a task-specific adapter, and on a real codebase, where sufficiency is a matter of degree rather than a constructed fact?

Three narrower lines follow. Population validity is open: the causal estimates rest on constructed workspaces, and the closest real-repository candidate went unaudited for leakage and fix-site resolution (C17). The hidden-type effect redacts a type rather than hiding it the way real code does, so transfer to generated stubs, vendored packages and compiled boundaries needs testing. The checker gate needs a rejection-precision

run over drafts that are not the defect’s own gold, and a design with repeated in-loop edits to exercise the after-every-edit channel.

Standing scope: static tooling does not reach dynamic behavior, incorrect annotations, **Any**-heavy boundaries or logic outside the checker, and the resolution answer holds only while the agent can cheaply read the type-bearing source. Real-repository substitution frequency and live-server latency are unmeasured, six four-seed files lack their seed-2/3 shards, and no artifact records a server version.

7. Conclusion

Semantic navigation added little on the dispatch suite tested, because the receiver’s type was readable in the source the agent opened; where that type was withheld, lookup helped, but only once the agent was framed to use it. A compact definition span costs less than a read only when it displaces one; pushing, instructing and self-election did not produce that displacement, and training did — producing a policy that goes looking when what it holds is inadequate and stops when it is not. A type check at the submission boundary, with a repair-and-resubmit loop, converted accepted bad completions into repaired ones on twelve seeded defect pairs.

References

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